# Sanaul Haque

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Assignment 202

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# How Much Can Timing Buy?

*A Quantity–Timing Benchmark of Peak Shaving on New Zealand's 2024 National Demand*

**Full Legal Name:** Yizhou Tang

**Institution:** Rangitoto College

**Category:** Junior

**Country:** New Zealand

**Registered Email:** tangrick0873@gmail.com

**Class/Grade/Year:** Year 13

**Submission Date:** 20 September 2026

## Research Problem/Significance

New Zealand electricity demand is usually highest on winter mornings and evenings. Heating is a big reason for these peaks. If more people use electricity for cars and process heat without any control, the peaks could get even higher [1], [12]. Meeting these peaks means we need extra generation and network capacity. Some of this need could be reduced by shifting demand, but the value of shifting depends on two things: how much load can be moved and how far in time it can be moved [1], [12].

Recent modelling by EECA shows about 1,700 to 1,900 MW of demand could be shifted. That is around 25% of the current peak. But the actual amount is limited by what customers will accept, the technology available, and the incentives [8], [10], [11]. Those models look at each technology and the limits in each sector. They do not show clearly how the amount of flexible load and the length of the time window work together, or which one is the main limit on a given day. This study tries to separate those two factors and give a simple reference point that other more detailed studies can compare against.

## Literature and Gap

Demand response is already known as a useful system resource. Strbac talks about its benefits and problems [1]. Palensky & Dietrich and Siano give classifications and survey results [2], [3]. Most studies use specific technical and time assumptions. They do not assume that all load can be moved. Gils separates theoretical potential from real participation [4]. Other papers explain how the time window for shifting decides what flexible loads can actually do [5], [6].

Using 2021 data, Kamana-Williams and Bishop say up to 69% of national electricity demand has some demand-response potential [7]. The EECA/Jacobs reports from 2026 have detailed data and a national flexibility signal [9], plus information about real-world limits [10], [11]. None of the New Zealand studies show a full grid of different flexible shares and different time windows on the same real national load curve. Making that grid only needs official data and a small linear program. It can give clear bounds and show which constraint is binding. The code will be open. This work is meant to sit beside the 2026 modelling, not replace it [8], [9].

## Research Question and Hypotheses

**Question:** Using New Zealand's 2024 half-hourly national offtake data, how much can daily peak demand be reduced if a fraction $\alpha$ of the demand in each half-hour can be moved by up to $\tau$ hours earlier or later in the same day? Which limit – the amount of flexible load, the length of the time window, or the total daily energy – stops further reduction across the different values of $\alpha$ and $\tau$?

$\alpha$ takes values 0, 0.05, …, 0.30. It is just a scenario number, not a measured adoption rate. $\tau$ takes values 0, 0.5, 1, 2, 4, 8 hours. A case with no time limit ($\tau = \infty$) is also included as a reference.

For each day d the fractional reduction is $r_d = (D_{max} - P_d^*) / D_{max}$, shown as $R_d = 100\, r_d$ (%). There are two simple limits:

$$r_d \le \alpha$$

$$r_d \le 1 - \bar{D} / D_{max}$$

When the time window has no limit, the best peak is $P_d^* = \max\{(1 - \alpha) D_{max}, \bar{D}\}$. The two limits meet when $\alpha$ equals $1 - \bar{D} / D_{max}$.

I expect:

**H1.** When τ is small, especially when α is larger, more days will be limited by the time window, so r_d will be smaller than both α and the energy limit.

**H2.** When τ gets bigger, the results will get closer to the no-limit case. Once the daily energy limit is reached, adding more flexible quantity does not help any more.

**H3.** Winter days usually have stronger morning and evening peaks, so they will need larger τ than summer days before they reach the energy limit.

The value of the method is that it uses real 2024 load shapes to find these boundaries. The closed-form answer cannot do that when τ is limited.

### Methodology

I will use the Wholesale → Volumes → Reconciliation files from the Electricity Authority. These are monthly files at half-hour resolution [13]. I will take the latest file for each month of 2024 and record the file names, preparation times and SHA-256 checksums. When FlowDirection is Offtake I will add the kWh values by TradingDate and TradingPeriod and change them to average megawatts by dividing by 500. Each day keeps its own number of periods (50 on 7 April, 46 on 29 September, 48 on all other days; 366 days in total).

D_t is the demand in period t. x_ts is the flexible load moved from period t to period s, where the distance between s and t is at most 2τ and both periods are inside the day. The linear program is:

$$\min P$$

$$\text{subject to}$$

$$\sum_s x_{ts} = \alpha D_t \quad \text{for every } t$$

$$L_s = (1 - \alpha) D_s + \sum_t x_{ts} \le P \quad \text{for every } s$$

$$x_{ts} \ge 0$$

This matches the simple theoretical-potential approach used in earlier papers [4]–[6]. It is simpler than the rolling-window method in [9]. There are 366 days × 7 values of α × 7 values of τ = 17,934 small problems. They can be solved with the open-source HiGHS solver. Code and results will be made public.

The main result is the distribution of daily R_d values (median, interquartile range and some percentiles) for each pair (α, τ). Secondary results include the highest optimised peak across the whole year and the average reduction on the top 10 and top 50 peak days. I will also show absolute MW reductions, seasonal patterns, and how much extra reduction comes from increasing α or τ. Each day-scenario will be classed as quantity-bound, energy-bound or timing-bound (or jointly bound if both limits are the same within tolerance).

For checking, I will make sure energy is conserved and no negative values appear. The unlimited-τ cases must match the closed-form answer. Solutions must also satisfy a simple interval inequality that follows from the limited reach of the flexible load. Results should get better (or stay the same) when α or τ is increased. I will also check for missing, duplicate or negative data and for correct handling of the two DST days.

For robustness I will also run a version that allows shifts across midnight and minimises the sum of daily peaks. The unlimited-τ case stays within-day only. Results will be split by season and by weekday/weekend. Because the peak-only solution is not unique, a second stage that minimises total movement while keeping the same peak will be used only to show one example schedule.

### Limitations, Ethics and Expected Value

The results are an upper bound under strong assumptions, not a forecast. The model assumes exactly $\alpha$ of every half-hour can be moved within $\pm\tau$. It ignores appliance deadlines, power limits, comfort, people opting out, prices and real adoption rates [10], [11]. National totals ignore network constraints. The objective only looks at peak size, not prices or emissions.

The study uses only public aggregate data, so there are no personal-data or human-subject issues. The main responsibility is to state the assumptions clearly, make the work reproducible, and not claim that the upper bound is what will actually happen.

The expected value is a simple reference layer that shows, for any assumed level of flexibility, how much national peak can be reduced and whether more flexible quantity or a wider time window gives more technical benefit on New Zealand's actual load shape. This can help later technology-specific and cost-benefit studies decide where to focus [8], [9].

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